

BIOLOGY

Majoring & Minorin

Introductory courses in the Biology Department can either serve as prerequisites for further study for the biology major/minor or satisfy the Natural Sciences area course requirement for General Education. We design advanced courses for students seeking more in-depth information across the areas of cell and molecular biology as well as ecology and evolutionary biology. In addition, some courses serve as requirements for students with specific vocational aims.

The Biology Department offers majors as BA and BS degrees as well as a minor. The Biology BA includes 32 credits of Biology courses, 8 credits of Chemistry and 4 credits of Statistics for a total of 44 credits. The Biology BS includes 36 credits of Biology, 12 credits of Chemistry, 8 credits of Mathematics/Computer Science, and 8 credits of Physics for a total of 64 credits. The Biology Department also supports the interdisciplinary major in Environmental Studies and supports several interdisciplinary minors.

By appropriate selection of course combinations, students can prepare for various options, such as entrance into graduate or professional schools (dental, medical, physician assistant, physical therapy, clinical laboratory scientist, nursing, optometry, pharmacy and veterinary) and acquisition of positions in industry, government, public health and teaching. Students should consult with their academic advisors and other members of the department for assistance in making proper course selections that will prepare them for their chosen career directions. The BS degree is recommended for students seeking entrance into professional schools, graduate schools or technician positions in industry, while the BA degree allows flexibility.

NOTE: Students must complete the first-year Biology sequence (Biology 50-173/171, 50-183/181) with a grade of C- or better before enrolling in either Biology 50-222 or 50-232. Students must complete the first-year biology sequence and either Biology 50-222 or 50-232 (see specific prerequisites for each advanced-level course) with a grade of C- or better before enrolling in any additional courses above the introductory level.

require a minimum of the introductory biology sequence and at least one Methods course as pre-requisites. The Biology Department recommends that a student consult with their advisor as to the appropriate advanced courses to take after completion of either one or both of the Methods courses (50-222 and 50-232). As for additional prerequisites, please refer to the course description for specifics. 900 level courses include capstone options, tutorials, honors, internships, research, and independent study.

Major in Biology (BA): 11-12 courses

- Two introductory courses:
 - BIO50-173 Molecular and Cellular Foundations of Biology, and
 - BIO50-183 Evolutionary and Ecological Foundations of Biology
- Two accompanying introductory laboratory courses:
 - BIO50-171 Inquiries in Biology, and
 - BIO50-181 Investigations in Biology
- Two intermediate courses:
 - BIO50-222 Methods in Ecology and Evolutionary Biology, and
 - BIO50-232 Methods in Cellular/Molecular Biology
- One advanced cellular/molecular course from:
 - BIO50-304 Special Topics (with permission of Department Chair when offered)
 - BIO50-354 Neurobiology
 - BIO50-394 Developmental Biology
 - BIO50-404 Cancer Biology
 - BIO50-464 Molecular Biology
 - BIO50-484 Microbiology
 - BIO50-494 Biology of Reproduction
 - BIO50-514 Cellular Physiology

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- BIO50-624 General Biochemistry II, or CHE51-624 General Biochemistry II
- One advanced ecology/evolutionary biology course from:
 - BIO50-304 Special Topics (with permission of Department Chair when offered)
 - BIO50-314 Genetics BIO50-334 Evolutionary Biology
 - BIO50-374 Human Genetics and Evolution
 - BIO50-384 Conservation Biology
 - BIO50-414 Plant Ecology
 - BIO50-434 Ecology
 - BIO50-444 Invertebrate Ecology
 - BIO50-474 Genetics, Genomics and Medicine
- Two additional courses (4 credits each) at 300-level or higher
- Capstone
 - Option 1:
 - BIO50-97X Research in Biology (3 total credits required), and BIO50-991 Biology Capstone Seminar OR
 - Option 2:
 - BIO50-994 Biology Capstone Seminar

Required supporting courses in the Biology major (BA): 5 courses

- CHE51-103 General Chemistry I
- CHE51-101 Chemical Methods and Techniques Lab I
- CHE51-203 General Chemistry II
- CHE51-201 Chemical Methods and Techniques Laboratory II
- MAT52-114 Introduction to Statistics

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- two introductory courses:
- BIO50-173 Molecular and Cellular Foundations of Biology, and
 - BIO50-183 Evolutionary and Ecological Foundations of Biology
- Two accompanying introductory laboratory courses:
 - BIO50-171 Inquiries in Biology, and
 - BIO50-181 Investigations in Biology
 - Two intermediate courses:
 - BIO50-222 Methods in Ecology and Evolutionary Biology, and
 - BIO50-232 Methods in Cellular/Molecular Biology
 - One advanced cellular/molecular course from:
 - BIO50-304 Special Topics (with permission of Department chair when offered)
 - BIO50-354 Neurobiology
 - BIO50-394 Developmental Biology
 - BIO50-404 Cancer Biology
 - BIO50-464 Molecular Biology
 - BIO50-484 Microbiology
 - BIO50-494 Biology of Reproduction
 - BIO50-514 Cellular Physiology
 - BIO50-534 Fundamentals of Immunology
 - BIO50-604 Principles of Biochemistry, or CHE51-604 Principles of Biochemistry
 - BIO50-624 General Biochemistry II, or CHE51-624 General Biochemistry II
 - One advanced ecology/evolutionary biology course from:
 - BIO50-304 Special Topics (with permission of Department chair when offered)
 - BIO50-314 Genetics
 - BIO50-334 Evolutionary Biology

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- BIO50-414 Plant Ecology
 - BIO50-434 Ecology BIO50-444
 - Invertebrate Ecology
 - BIO50-474 Genetics, Genomics and Medicine
- Three additional courses (4 credits each) at 300-level or higher
- Capstone
 - Option 1:
 - BIO50-97X Research in Biology (3 total credits required), and
 - BIO50-991 Biology Capstone Seminar OR
 - Option 2:
 - BIO50-994 Biology Capstone Seminar

Required supporting courses in the Biology major (BS): 10 courses

- CHE51-103 General Chemistry I
- CHE51-101 Chemical Methods and Techniques Laboratory I
- CHE51-203 General Chemistry II
- CHE51-201 Chemical Methods and Techniques Laboratory II
- CHE51-313 Organic Chemistry I
- CHE51-311 Organic Chemistry Laboratory Techniques
- MAT52-114 Introduction to Statistics
- MAT52-164 Modern Calculus I
- PHY53-154 Fundamentals of Physics I
- PHY53-164 Fundamentals of Physics II

Note: Excluding the Capstone seminar, at least two of the four total 300-level or above courses for the BA and at least three of the five total 300-level or above courses for the BS must have a

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- Two introductory courses:
 - BIO50-173 Molecular and Cellular Foundations of Biology, and
 - BIO50-183 Evolutionary and Ecological Foundations of Biology
- Two accompanying introductory laboratory courses:
 - BIO50-171 Inquiries in Biology, and
 - BIO50-181 Investigations in Biology
- At least one intermediate course:
 - BIO50-222 Methods in Ecology and Evolution, or
 - BIO50-232 Methods in Cell and Molecular Biology
- 8 credits of upper-level biology, 300-level or above (at least one course with lab component)

See the Education Department for information regarding teacher certification in Biology.

GEORGETOWN, TEXAS

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